Project Initialization and Planning Phase

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Title	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal report aims to transform educational assistance using generative AI models, boosting learning efficiency and student engagement. It tackles traditional study content complexities, promising faster concept understanding, automated practice quiz generation, and personalized learning paths. Key features include Google Gemini 1.5 Pro integration, local lightweight AI models, real-time Q&A, and content summarization.

Project Overview	
Objective	The primary objective is to revolutionize the self-learning process by implementing advanced Generative AI techniques, ensuring faster concept clarification, automated study material processing, and customized learning recommendations.
Scope	The project comprehensively assesses and enhances the learning workflow, incorporating Google Gemini 1.5 Pro API and LaMini-Flan-T5 models for a lightweight, accessible, and interactive AI learning assistant.
Problem Statement	
Description	Complex study content, lack of personalized tutorship, and high resource requirements of heavy AI tools create barriers for students trying to understand topics efficiently.
Impact	Solving these issues will result in improved study productivity, reduced learning time, and enhanced concept retention, contributing to student success and academic satisfaction.
Proposed Solution	
Approach	Employing Generative AI and NLP techniques to analyze educational text, simplify complex ideas, generate interactive quizzes, and recommend tailored learning roadmaps.
Key Features	• Text Summarization: Summarizes lengthy educational articles and documents into actionable insights. • Personalized Learning Paths: Provides structured, step-by-step learning roadmaps for specified topics.

	• Dual Model Support: Uses Google Gemini API for cloud intelligence and LaMini-Flan-T5 for fast local CPU processing.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard Dual/Quad-Core CPU / Apple Silicon (Mac M1 Compatible)
Memory	RAM specifications	8 GB / 16 GB RAM
Storage	Disk space for data, models, and logs	20 GB SSD (Model weights & project environment)
Software		
Frameworks	Python frameworks	FastAPI, Jinja2 Templating
Libraries	Additional libraries	google-generative ai, transformers, torch, uvicorn, pydantic
Development Environment	IDE	VS Code / PyCharm / Jupyter Notebook
Data		
Data	Source, size, format	Custom User Text Inputs, Prompt Templates, Structured JSON Output formats